IDENTIFYING INFORMATION:

NAME: Brekken, Ted

ORCID iD: https://orcid.org/0000-0002-3093-108X

POSITION TITLE: Professor

PRIMARY ORGANIZATION AND LOCATION: Oregon State University, Corvallis, Oregon, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of Minnesota, Minneapolis, MN, USA	PHD	09/2002 - 06/2005	Electrical Engineering
University of Minnesota, Minneapolis, MN, USA	MS	09/2000 - 06/2002	Electrical Engineering
University of Minnesota, Minneapolis, MN, USA	BS	09/1994 - 12/1999	Electrical Engineering

Appointments and Positions

2006 - present Professor, Oregon State University, Corvallis, Oregon, United States
2026 - present Director, Pacific Marine Energy Center, Corvallis, Oregon, United States
2018 - present Professor, Oregon State University, Corvallis, OR, USA
2012 - 2018 Associate Professor, Oregon State University, Corvallis, OR, USA
2006 - 2012 Assistant Professor, Oregon State University, Corvallis, OR, USA
2005 - 2006 Post-Doc, University of Minnesota, Minneapolis, MN, USA
2005 - 2006 Adjunct Professor, University of St. Thomas, St. Paul, MN, USA
2004 - 2005 Fulbright Scholar, Norwegian University of Science and Technology, Trondheim, Not Applicable, N/A, Norway

Products

Products Most Closely Related to the Proposed Project

1. Shifat T, Coe R, Bacelli G, Brekken T. Constrained Pseudo-PI Linear Control of a Wave Energy Converter via Model Predictive Control. IFAC-PapersOnLine. 2024; 58(20):398-403. Available from: https://linkinghub.elsevier.com/retrieve/pii/S240589632401841X DOI: 10.1016/j.ifacol.2024.10.086
2. Barton S, Brekken T, Cao Y. Extending Power Electronic Converter Lifetime in Marine Hydrokinetic Turbines with Reinforcement Learning. Applied Sciences. 2025 February 26; 15(5):2512-. Available from: https://www.mdpi.com/2076-3417/15/5/2512 DOI: 10.3390/app15052512
3. Kim I, Brekken T, Yim S, Johnson B, Cao Y, Chandran P. Real-Time Wave Energy Converter

Control Using Instantaneous Frequency. Applied Sciences. 2025 April 28; 15(9):4889-. Available from: https://www.mdpi.com/2076-3417/15/9/4889 DOI: 10.3390/app15094889

4. Shifat T, Coe R, Bacelli G, Brekken T. Real-Time Sea State Estimation for Wave Energy Converter Control via Machine Learning. Applied Sciences. 2025 May 21; 15(10):5772-. Available from: https://www.mdpi.com/2076-3417/15/10/5772 DOI: 10.3390/app15105772
5. Reinforcement Learning Control for Enhancing Marine Hydrokinetic Turbine Energy Generation. 2024 AMERICAN CONTROL CONFERENCE, ACC 2024. 2024. wosuid: WOS:001310893801003

Other Significant Products, Whether or Not Related to the Proposed Project

1. Haider Ali, Stark Will, Brekken Ted KA. Electric hot water heater primary frequency control. 2018 IEEE Energy Conversion Congress and Exposition (ECCE); 2018; c2018.
2. Zadeh L, Brekken T, Fern A, Shahbaz A. Hardware in the Loop Wave Energy Converter Control Under Control Faults and Model Mismatch. IEEE Transactions on Sustainable Energy. 2024; 15(1):13-22. Available from: https://ieeexplore.ieee.org/document/10122770/ DOI: 10.1109/TSTE.2023.3272537
3. Zadeh L, Haider A, Brekken T. Bayesian Actor-Critic Wave Energy Converter Control With Modeling Errors. IEEE Transactions on Sustainable Energy. 2023; 14(1):3-11. Available from: https://ieeexplore.ieee.org/document/9770354/ DOI: 10.1109/TSTE.2022.3172426
4. Ling B, Bosma B, Brekken T. Experimental Validation of Model Predictive Control Applied to the Azura Wave Energy Converter. IEEE Transactions on Sustainable Energy. 2020 October; 11(4):2284-2293. Available from: https://ieeexplore.ieee.org/document/8903474/ DOI: 10.1109/TSTE.2019.2953868
5. Ni Y, Seki A, Bosma B, Brekken T, Robertson B, Schellenberg A, Lomonaco P, Simpson B. A Robotic Actuation Approach for 6-DOF Force Control in Hydrodynamic Real-Time Hybrid Simulation *. 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE). 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE); ; Los Angeles, CA, USA. IEEE; c2025. Available from: https://ieeexplore.ieee.org/document/11164156/ DOI: 10.1109/CASE58245.2025.11164156

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Brekken, Ted in SciENcv on 2026-10-15 18:28:19